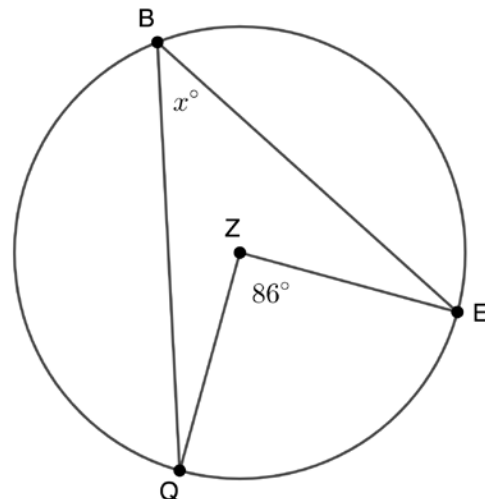


Geometry
Unit 10
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. Circle Z is shown where $\angle EZQ$ is a central angle, $\angle EBQ$ is inscribed in the circle, and $m\angle EZQ = 86^\circ$. Find $m\angle EBQ$.

$$m\angle EBQ = \frac{86}{2} = 43^\circ$$

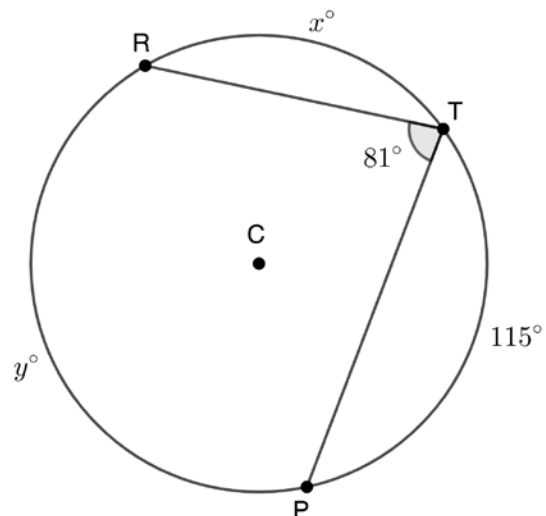


For questions 2-3, Circle C has points R , T , and P on the circle. $\angle RTP$ is inscribed in Circle C measuring 81° , and the

$$m\widehat{TP} = 115^\circ.$$

2. Find the $m\widehat{RP}$.

$$m\widehat{RP} = 2 \cdot 81 = 162^\circ$$



3. Find the $m\widehat{RT}$.

$$m\widehat{RT} = 360 - 115 - 162 = 83^\circ$$

For questions 4-5, points O , P , Q , and R lie on Circle S . $\angle PQR$ is inscribed in the circle, the $m\angle PQR = (5x + 22)^\circ$, and the $m\widehat{ROP} = 194^\circ$.

4. Find the $m\angle PQR$.

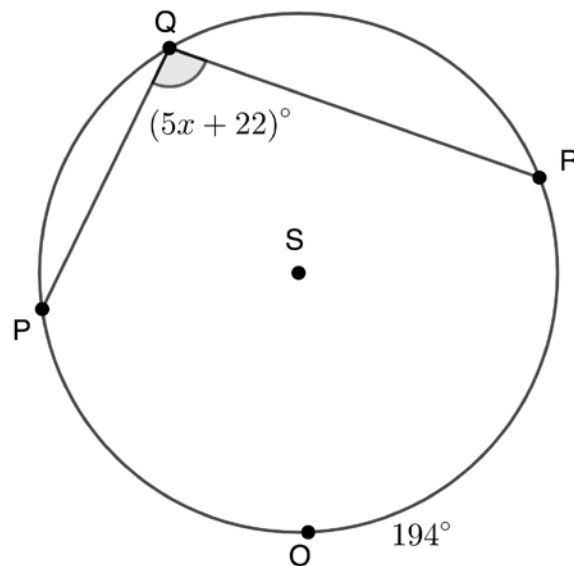
$$m\angle PQR = \frac{194}{2} = 97^\circ$$

5. Find the value of x .

$$5x + 22 = 97$$

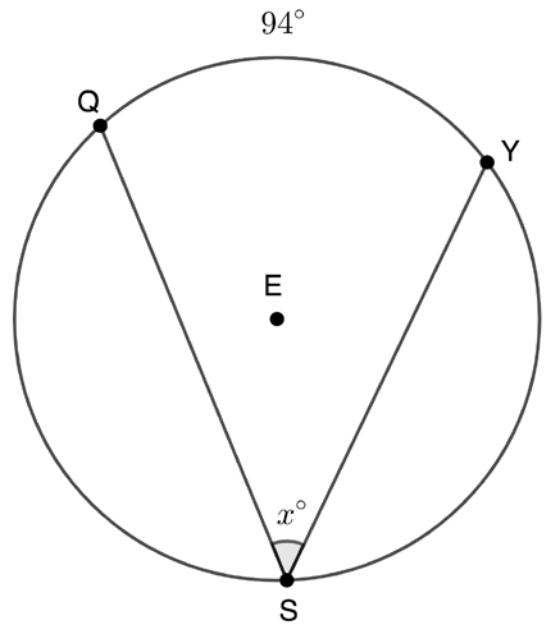
$$5x = 75$$

$$x = 15$$



6. Circle E is shown, where points Q , Y , and S are on the circle. $\angle QSY$ is inscribed in Circle E with $m\widehat{QY} = 94^\circ$. Find the value of x .

$$x = \frac{94}{2} = 47$$



For questions 7-10, Circle Q is shown with inscribed angle $\angle LKJ$ and diameter $\overline{JL}$. The $m\widehat{KJ} = (12x - 14)^\circ$ and the $m\widehat{KL} = (8x - 6)^\circ$.

7. Find the value of x .
 $12x - 14 + 8x - 6 = 180$
 $20x - 20 = 180$
 $20x = 200$
 $x = 10$

8. Find the $m\angle KJL$.
 $\widehat{KL} = 8(10) - 6 = 74^\circ$
 $m\angle KJL = \frac{74}{2} = 37^\circ$

9. Find the $m\angle KLJ$.
 $\widehat{KJ} = 12(10) - 14 = 106^\circ$
 $m\angle KLJ = \frac{106}{2} = 53^\circ$

10. Find the $m\widehat{KJ}$.
 $m\widehat{KJ} = 12(10) - 14 = 106^\circ$

